in $\mathscr{L}^\sharp_\lambda$ is given here by Eq. (29.3.9), but with the last term absent:

$$\mathscr{B}_\lambda = Y f_\lambda(\Phi, \exp(2i\pi T)) + \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{B\beta B}\, \ell_{\lambda AB}(\Phi, \exp(2i\pi T))\,. \tag{29.3.32}$$

Because f_λ may depend on T, we cannot now conclude that it is equal to the bare superpotential, but only that it depends linearly on whatever coupling coefficients and masses appear linearly in the bare superpotential. In particular, *if there is no superpotential to begin with, then none is generated by non-perturbative effects.*

To go further, we need to make use of the anomalous chiral symmetry under a $U(1)$ transformation of all the Φ_n. In order for the whole theory to be invariant under this symmetry, it is necessary to introduce a separate external left-chiral superfield Y_r for the terms in the bare superpotential of order r in the Φ_n. Then the theory is invariant under the combined transformations

$$\Phi_n \to e^{i\varphi}\Phi_n\,, \qquad T \to T + C_2\varphi/\pi\,, \qquad Y_r \to e^{-ir\varphi} Y_r\,. \tag{29.3.33}$$

This symmetry tells us that a term in $\mathscr{B}_\lambda$ that is of order $\mathscr{N}_r$ in the coefficients of the term in the superpotential of order r in Φ and proportional to $\exp(2ai\pi T)$ must be of an order $\mathscr{N}_\Phi$ in Φ, given by

$$\mathscr{N}_\Phi = \sum_r r\mathscr{N}_r - 2C_2 a\,. \tag{29.3.34}$$

The coefficients $\ell_{\lambda AB}$ of the terms in $\mathscr{B}_\lambda$ that are quadratic in W are shown in Eq. (29.3.32) to be independent of the parameters in the superpotential for $C_1 = C_2$, so in this case Eq. (29.3.34) becomes

$$\mathscr{N}_\Phi = -2C_2 a\,. \tag{29.3.35}$$

Thus there can be no terms in $\ell_{\lambda AB}$ of positive order in the Φ_r, and any term in $\ell_{\lambda AB}$ that is independent of the Φ_r must be independent of T. These Φ-independent terms in $\ell_{\lambda AB}$ are therefore again just the one-loop contribution to the running coupling parameter τ_λ.

The effective superpotential is shown by Eq. (29.3.32) to be linear in the parameters in the superpotential, so all of its terms have just one $\mathscr{N}_r = 1$ and the others zero. For such a term Eq. (29.3.34) gives

$$\mathscr{N}_\Phi = r - 2C_2 a\,. \tag{29.3.36}$$

A term in the effective superpotential with $\mathscr{N}_\Phi$ powers of Φ can therefore only arise from terms in the bare superpotential with $r \geq \mathscr{N}_\Phi$ powers of Φ. The terms with $r = \mathscr{N}_\Phi$ have $a = 0$, so they are given by the tree approximation as just the bare superpotential. The only other terms are non-perturbative corrections with $r > \mathscr{N}_\Phi$. Such a non-perturbative